Math for CS & AI: Homework 4

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Problem 2.24

Let $\mathrm{Perm}(2n)$ be the set of all $(2n)!$ permutations of $[2n]$. For any $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_{2n}) \in \mathrm{Perm}(2n)$, we say that a pair of positions (i, j) with $i < j$ is an inversion of σ if $\sigma_i > \sigma_j$. For example, the permutation $(2, 4, 1, 3)$ has three inversions: $(1, 3)$, $(2, 3)$, and $(2, 4)$.

For any $\sigma \in \mathrm{Perm}(2n)$, let $f(\sigma)$ be the permutation obtained from σ by sorting the sublist of odd positions in increasing order. Let X be the number of inversions of $f(\sigma)$ when σ is chosen uniformly at random from $\mathrm{Perm}(2n)$. Compute $\mathbb{E}[X]$.

Answer:

By the linearity of expectation, we have:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{1 \le i < j \le 2n} \mathbb{1}_{[f(\sigma)_i > f(\sigma)_j]}\right] = \sum_{1 \le i < j \le 2n} \Pr[f(\sigma)_i > f(\sigma)_j].$$

Uniform random permutations are symmetric, which means that the probability of a particular order relationship among elements in a set of positions is unaffected by the elements in other positions.

Let $S_{\mathrm{odd}} = \{\sigma_{2i-1} \mid 1 \le i \le n\}$ denote the set of elements in the odd positions of the initial permutation σ.

For each pair with $1 \le i < j \le 2n$, we discuss the parity of i and j in four cases:

- If both i and j are odd, then by the definition of $f(\sigma)$, the elements in odd positions are sorted in increasing order, so $f(\sigma)_i < f(\sigma)_j$. This implies $\Pr[f(\sigma)_i > f(\sigma)_j] = 0$.
- If both i and j are even, then the elements in even positions remain unchanged from the original permutation σ. By the symmetry of uniform random permutations, $\Pr[f(\sigma)_i > f(\sigma)_j] = \Pr[f(\sigma)_i < f(\sigma)_j] = \frac{1}{2}$.
- If i is odd and j is even, assume $i = 2i_0 - 1$, which means $f(\sigma)_i$ is the i_0-th smallest element of S_{odd}. The element $f(\sigma)_j = \sigma_j$ and the n elements of S_{odd} form a set of $n+1$ distinct elements with uniform relative order. Thus, $\Pr[f(\sigma)_i > f(\sigma)_j] = \frac{i_0}{n+1}$.
- If i is even and j is odd, assume $j = 2j_0 - 1$, which means $f(\sigma)_j$ is the j_0-th smallest element of S_{odd}. By the same symmetry argument as above, we have $\Pr[f(\sigma)_i > f(\sigma)_j] = \frac{n - j_0 + 1}{n+1}$.

Summing the contributions from all cases and simplifying, we obtain:

$$\begin{aligned}\mathbb{E}[X] &= \frac{n(n-1)}{2} \cdot \frac{1}{2} + \sum_{i_0=1}^{n} \frac{i_0}{n+1} \cdot (n - i_0 + 1) + \sum_{j_0=1}^{n} \frac{n - j_0 + 1}{n+1} \cdot (j_0 - 1) \\ &= \frac{n(7n-1)}{12}.\end{aligned}$$

Problem 2.25

Let $\beta = 0.7$. For every $n \geq 10$, show that there exists a probability space $P = (\Omega, p)$ and n random variables $X_1, X_2, \ldots, X_n$ on P such that:

- $\Pr[X_i > X_{i+1}] \geq \beta$ for all $i = 1, 2, \ldots, n-1$;
- $\Pr[X_n > X_1] \geq \beta$.

Furthermore, for each $n \geq 10$, determine the largest possible value of β for which such a construction is possible.

Answer:

Let event A_i denote the event that $X_i > X_{(i \bmod n)+1}$.

For each $n \geq 10$, the maximum valid value of β is $\beta_{\max} = 1 - \frac{1}{n}$. We first construct a probability space and random variables achieving this bound, then prove that no larger β is possible.

Consider a probability space $P = (\Omega, p)$ where $\Omega = \{\omega_i \mid 1 \leq i \leq n\}$, and for each $\omega_i \in \Omega$, the probability mass function is $p(\omega_i) = \frac{1}{n}$.

For each outcome ω_i, we define the random variables X_j as:

$$X_j = \begin{cases} -j, & \text{if } 1 \leq j \leq i, \\ n - j, & \text{if } i < j \leq n. \end{cases}$$

Under this construction, the event A_j occurs if and only if $i \neq j$. For each $n \geq 10$, this gives $\Pr[A_j] = 1 - \frac{1}{n} \geq 0.7$, which satisfies the problem's requirement for $\beta = 0.7$.

We now prove that $1 - \frac{1}{n}$ is the maximum possible value of β. Assume for contradiction that there exists some $\beta > 1 - \frac{1}{n}$ satisfying the problem's conditions. For each i, we have $\Pr[\bar{A}_i] = 1 - \Pr[A_i] < \frac{1}{n}$. Using the union bound, we obtain:

$$\Pr\left[\bigcup_{i=1}^{n} \bar{A}_i\right] \leq \sum_{i=1}^{n} \Pr[\bar{A}_i] < 1.$$

The complement event of $\bigcup_{i=1}^{n} \bar{A}_i$ is $\bigcap_{i=1}^{n} A_i$, which means

$$\Pr\left[\bigcap_{i=1}^{n} A_i\right] = 1 - \Pr\left[\bigcup_{i=1}^{n} \bar{A}_i\right] > 0.$$

However, these n events cannot occur simultaneously, as it would imply the cyclic inequality $X_1 > X_2 > \cdots > X_n > X_1$, a logical contradiction. Thus, no such $\beta > 1 - \frac{1}{n}$ can exist.

In conclusion, for each $n \geq 10$, the largest possible valid value of β is $1 - \frac{1}{n}$.

Problem 2.26

Let X and Y be two random variables and $f : \mathbb{R} \to \mathbb{R}$ be a function. Simplify the following two expressions that might confuse you:

(a) $\mathbb{E}\left[\mathbb{E}\left[X \mid f(Y)\right] \mid Y\right]$;

(b) $\mathbb{E}\left[\mathbb{E}\left[X \mid Y\right] \mid f(Y)\right]$.

Answer:

Both expressions are conditional expectations, which are functions from the sample space Ω to $\mathbb{R}$.

For the expression (a), by the definition of conditional expectation, we have:

$$\begin{aligned}
\mathbb{E}\left[\mathbb{E}[X \mid f(Y)] \mid Y = Y(\omega)\right] &= \sum_t \mathbb{E}[X \mid f(Y) = t] \cdot \Pr[f(Y) = t \mid Y = Y(\omega)] \\
&= \mathbb{E}[X \mid f(Y) = f(Y(\omega))] \cdot \Pr[f(Y) = f(Y(\omega)) \mid Y = Y(\omega)] \\
&= \mathbb{E}[X \mid f(Y) = f(Y(\omega))],
\end{aligned}$$

which means expression (a) is equivalent to $\mathbb{E}[X \mid f(Y)]$.

Let $S_\omega = \{y \mid f(y) = f(Y(\omega))\}$ denote the set of values y that $f(y) = f(Y(\omega))$.

For the expression (b), by the law of total expectation and the definition of conditional probability, we have:

$$\begin{aligned}
\mathbb{E}\left[\mathbb{E}[X \mid Y] \mid f(Y) = f(Y(\omega))\right] &= \sum_{y \in S_\omega} \mathbb{E}[X \mid Y = y] \cdot \Pr[Y = y \mid f(Y) = f(Y(\omega))] \\
&= \sum_{y \in S_\omega} \mathbb{E}[X \mid Y = y] \cdot \frac{\Pr[Y = y]}{\Pr[f(Y) = f(Y(\omega))]} \\
&= \frac{1}{\Pr[f(Y) = f(Y(\omega))]} \sum_{y \in S_\omega} \sum_x x \cdot \Pr[X = x \mid Y = y] \cdot \Pr[Y = y] \\
&= \frac{1}{\Pr[f(Y) = f(Y(\omega))]} \sum_x x \sum_{y \in S_\omega} \Pr[X = x, Y = y] \\
&= \frac{1}{\Pr[f(Y) = f(Y(\omega))]} \sum_x x \cdot \Pr[X = x, f(Y) = f(Y(\omega))] \\
&= \sum_x x \cdot \Pr[X = x \mid f(Y) = f(Y(\omega))] \\
&= \mathbb{E}[X \mid f(Y) = f(Y(\omega))].
\end{aligned}$$

which means expression (b) is also equivalent to $\mathbb{E}[X \mid f(Y)]$.

Problem 2.27

Let $\Omega = \{2^k : k \in \{0, 1, 2, \ldots, m\}\}$, where $m = 10^5$. Alice chooses a random number $X \in \Omega$ uniformly at random. Since Alice is very rich, she puts X RMB in one envelope and $2X$ RMB in the other envelope. Bob chooses one envelope uniformly at random and opens it to see the amount of money inside. Bob can then decide to keep the money or switch to the other envelope.

Let Y be the amount of money inside Bob's envelope, Z be the amount of money inside the other envelope, and M be the final amount of money Bob gets according to his decision. Bob wants to maximize the expected value of M, i.e., $\mathbb{E}[M \mid Y]$.

Bob thinks for a while and then concludes that he should switch to the other envelope no matter what Y he sees. The reason is as follows:

- There is equal probability that $Y = X$ and $Y = 2X$.
- If $Y = X$, then $Z = 2Y$. If $Y = 2X$, then $Z = Y/2$.
- By the law of total expectation, $\mathbb{E}[Z \mid Y] = \frac{1}{2} \cdot 2Y + \frac{1}{2} \cdot \frac{Y}{2} = 1.25Y > Y$.
- Therefore, setting $M = Z$ leads to a higher expected value of M.

Is Bob's conclusion correct? If yes, just say yes. If no, compute $\mathbb{E}[Z \mid Y]$.

Answer:

Bob's conclusion is incorrect. The error in Bob's reasoning is his incorrect assumption that the events $Y = X$ and $Y = 2X$ occur with equal conditional probability for all values of Y; this assumption fails for the extreme values of Y. The correct expression for $\mathbb{E}[Z \mid Y]$ is:

$$\mathbb{E}[Z \mid Y] = \begin{cases} 2Y, & \text{if } Y = 2^0, \\ \frac{1}{2}Y, & \text{if } Y = 2^{m+1}, \\ \frac{5}{4}Y, & \text{otherwise.} \end{cases}$$

Problem 2.28

Let $X_1, X_2, \ldots, X_n$ be n i.i.d. random variables that are drawn uniformly at random from $\{-1, 1\}$. Are the following random variables also i.i.d.? Prove your answer.

$$Y_1 = X_1,\ Y_2 = X_1X_2,\ \ldots,\ Y_n = X_1X_2\cdots X_n.$$

Answer:

Yes, the random variables $Y_1, Y_2, \ldots, Y_n$ are i.i.d. We prove this result below.

Since $Y_1 = X_1$, we have that Y_1 is uniformly distributed over $\{-1, 1\}$, with $\Pr[Y_1 = 1] = \Pr[Y_1 = -1] = \frac{1}{2}$. We proceed by induction to show that for all $1 \le k \le n$, Y_k is uniformly distributed over $\{-1, 1\}$, and that $Y_1, \ldots, Y_k$ are mutually independent.

Assume that for some $1 \le i < n$, $Y_1, \ldots, Y_i$ are mutually independent, with each Y_j $(1 \le j \le i)$ uniformly distributed over $\{-1, 1\}$. Note that $Y_{i+1} = Y_i \cdot X_{i+1}$, where X_{i+1} is independent of $Y_1, \ldots, Y_i$: since $Y_1, \ldots, Y_i$ are functions of $X_1, \ldots, X_i$ and $X_1, \ldots, X_n$ are i.i.d., X_{i+1} is independent of all $Y_1, \ldots, Y_i$. By the law of total probability:

$$\begin{aligned}
\Pr[Y_{i+1} = 1] &= \Pr[Y_i = 1, X_{i+1} = 1] + \Pr[Y_i = -1, X_{i+1} = -1] \\
&= \Pr[Y_i = 1]\Pr[X_{i+1} = 1] + \Pr[Y_i = -1]\Pr[X_{i+1} = -1] \\
&= \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \\
&= \frac{1}{2}.
\end{aligned}$$

Similarly, $\Pr[Y_{i+1} = -1] = 1 - \Pr[Y_{i+1} = 1] = \frac{1}{2}$, so Y_{i+1} is also uniformly distributed over $\{-1, 1\}$.

We next verify that Y_{i+1} is independent of $Y_1, \ldots, Y_i$. For any fixed values $a_1, \ldots, a_i, a_{i+1} \in \{-1, 1\}$:

$$\begin{aligned}
\Pr[Y_1 = a_1, \ldots, Y_i = a_i, Y_{i+1} = a_{i+1}] &= \Pr[Y_1 = a_1, \ldots, Y_i = a_i, Y_iX_{i+1} = a_{i+1}] \\
&= \Pr[Y_1 = a_1, \ldots, Y_i = a_i, X_{i+1} = a_{i+1}a_i] \\
&= \Pr[Y_1 = a_1, \ldots, Y_i = a_i] \cdot \Pr[X_{i+1} = a_{i+1}a_i] \\
&= \left(\prod_{j=1}^{i} \Pr[Y_j = a_j]\right) \cdot \frac{1}{2} \\
&= \prod_{j=1}^{i+1} \Pr[Y_j = a_j].
\end{aligned}$$

By induction, all $Y_1, Y_2, \ldots, Y_n$ are mutually independent and each is uniformly distributed over $\{-1, 1\}$, meaning they are i.i.d. random variables.

Problem 2.29

For any two binary vectors $x, y \in \{0,1\}^d$, define the Hamming distance as

$$\text{dist}_H(x,y) = \sum_{i=1}^{d} \mathbb{1}_{[x_i \neq y_i]}.$$

Construct a family $\mathcal{H}$ of hash functions $h : \{0,1\}^d \to \{0,1\}$ such that for any two binary vectors $x, y \in \{0,1\}^d$, when h is sampled uniformly at random from $\mathcal{H}$,

$$\Pr[h(x) = h(y)] = 1 - \frac{1}{d}\,\text{dist}_H(x,y).$$

Answer:

We construct the hash family $\mathcal{H}$ as follows: let $\mathcal{H} = \{h_i \mid 1 \leq i \leq d\}$, where each hash function h_i is defined by $h_i(x) = x_i$ for all $x \in \{0,1\}^d$. Let A_i denote the event that the hash function h_i is selected when sampling uniformly at random from $\mathcal{H}$.

By the law of total probability, we have:

$$\begin{aligned}
\Pr[h(x) = h(y)] &= \sum_{i=1}^{d} \mathbb{1}_{[h_i(x)=h_i(y)]} \cdot \Pr[A_i] \\
&= \sum_{i=1}^{d} \mathbb{1}_{[x_i=y_i]} \cdot \frac{1}{d} \\
&= \frac{1}{d} \sum_{i=1}^{d} (1 - \mathbb{1}_{[x_i \neq y_i]}) \\
&= 1 - \frac{1}{d}\,\text{dist}_H(x,y).
\end{aligned}$$

Problem 3.1

Is it possible that for some random variable X that takes non-negative integer values, Markov's inequality (Theorem 3.1) only loses by a constant factor from being tight for all integers $a > 0$? That is, is there a random variable X and a constant $c > 0$ such that the following inequality holds for all integers $a > 0$?

$$\frac{c\mathbb{E}[X]}{a} \leq \Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

Answer:

For the degenerate random variable $X \equiv 0$ (taking the value 0 with probability 1) and any constant $c > 0$, for every integer $a > 0$, we have $\Pr[X \geq a] = 0$ and $\mathbb{E}[X] = 0$, which satisfies the required inequality.

However, for all non-trivial cases where $\Pr[X > 0] > 0$ (and thus $\mathbb{E}[X] > 0$ for non-negative X), no such random variable X and constant $c > 0$ exist. We prove this by contradiction below.

Assume for contradiction that there exists a non-negative integer-valued random variable X with $\mathbb{E}[X] > 0$ and a constant $c > 0$ satisfying the given inequality for all integers $a > 0$. By the tail-sum formula, we have:

$$\begin{aligned}\mathbb{E}[X] &= \sum_{a \geq 1} \Pr[X \geq a] \\ &\geq \sum_{a \geq 1} \frac{c\mathbb{E}[X]}{a}.\end{aligned}$$

Since $\mathbb{E}[X] > 0$, we divide both sides of the inequality by $\mathbb{E}[X]$ and rearrange it, which gives:

$$\frac{1}{c} \geq \sum_{a \geq 1} \frac{1}{a}.$$

However, the right-hand side of the inequality is the harmonic series, which diverges to $+\infty$. This leads to a contradiction, as no finite constant $c > 0$ can satisfy the inequality.

Therefore, the only random variable satisfying the condition is the degenerate zero random variable, and no non-trivial such random variable exists.

Problem 3.2

A function $f : \mathbb{R} \to \mathbb{R}$ is called convex if for all $x, y \in \mathbb{R}$ and $t \in [0, 1]$,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y).$$

A convex function is always continuous. It may not be differentiable, but for any convex function f, the left-derivative $f'_-(x) := \lim_{h\to 0^-} \frac{f(x+h)-f(x)}{h}$ and the right-derivative $f'_+(x) := \lim_{h\to 0^+} \frac{f(x+h)-f(x)}{h}$ exist at every point $x \in \mathbb{R}$.

(a) Prove the following statement: If $f : \mathbb{R} \to \mathbb{R}$ is a convex function, then for any $y \in \mathbb{R}$, there exists a factor $\alpha \in \mathbb{R}$ such that for all $x \in \mathbb{R}$

$$f(x) \geq f(y) + \alpha(x - y).$$

That is, the function f is lower bounded by a line with slope α that passes through the point $(y, f(y))$.

(b) Let X be a random variable and $f : \mathbb{R} \to \mathbb{R}$ be a convex function. Prove Jensen's inequality for random variables:

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

Answer:

For part (a), we rearrange the definition of convex functions and obtain:

$$f(y + t(x - y)) - f(y) \leq t\left[f(x) - f(y)\right].$$

For the case that $x > y$, we take the limit $t \to 0^+$ and divide both sides of the inequality by $t(x - y)$, which is positive:

$$\lim_{t\to 0^+} \frac{f(y + t(x - y)) - f(y)}{t(x - y)} \leq \frac{f(x) - f(y)}{x - y},$$

where the left-hand side of the inequality is the right-derivative $f'_+(y)$ at y. Rearranging the inequality again, we obtain:

$$\forall x > y,\ f(x) \geq f(y) + f'_+(y)(x - y).$$

For the case that $x < y$, we take the limit $t \to 0^+$ and divide both sides of the inequality by $t(x - y)$, which is negative (thus reversing the inequality direction):

$$\lim_{t\to 0^+} \frac{f(y + t(x - y)) - f(y)}{t(x - y)} \geq \frac{f(x) - f(y)}{x - y},$$

where the left-hand side of the inequality is the left-derivative $f'_-(y)$ at y. Rearranging the inequality again, we obtain:

$$\forall x < y,\ f(x) \geq f(y) + f'_-(y)(x - y).$$

For convex functions, the standard property $f'_-(y) \leq f'_+(y)$ holds for all $y \in \mathbb{R}$. Combining both cases, we choose $\alpha = f'_+(y)$ (or $\alpha = f'_-(y)$), which satisfies the required inequality for all $x \in \mathbb{R}$.

Problem 3.2 continued on next page...

For part (b), we apply the result from part (a) with $y = \mathbb{E}[X]$. Let α be a slope satisfying the lower bound inequality for this fixed y. Then for every possible value of the random variable X, we have:

$$f(X) \geq f(\mathbb{E}[X]) + \alpha(X - \mathbb{E}[X]).$$

Taking the expectation of both sides and using the linearity of expectation, we obtain:

$$\begin{aligned}\mathbb{E}[f(X)] &\geq \mathbb{E}[f(\mathbb{E}[X]) + \alpha(X - \mathbb{E}[X])] \\ &= f(\mathbb{E}[X]) + \alpha\mathbb{E}[X - \mathbb{E}[X]] \\ &= f(\mathbb{E}[X]).\end{aligned}$$